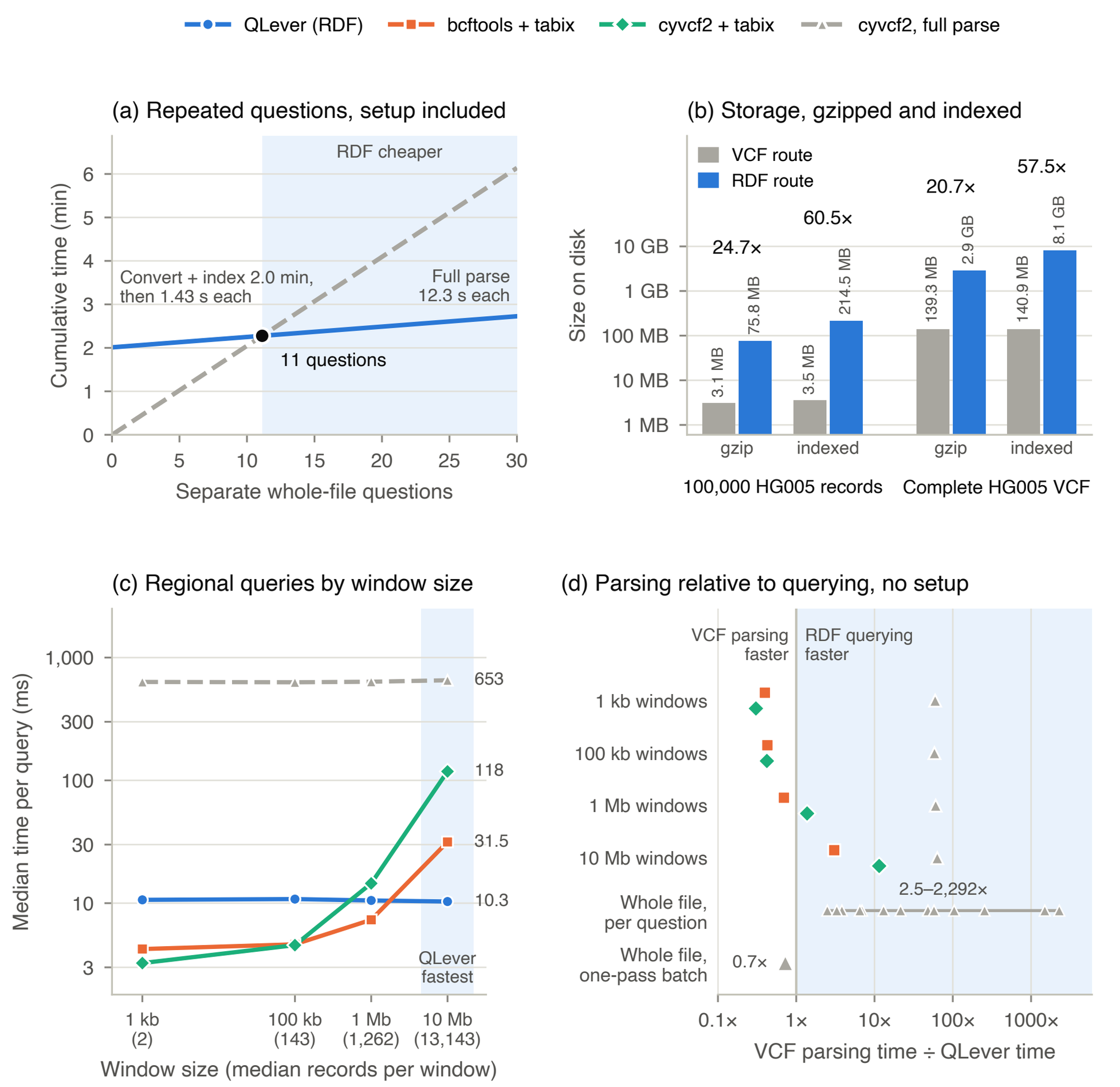
QLever (RDF)
bcftools + tabix
cyvcf2 + tabix
cyvcf2, full parse
(a) Repeated questions, setup included
RDF cheaper
Convert + index 2.0 min,
then 1.43 s each
Full parse
12.3 s each
11 questions
Cumulative time (min)
Separate whole-file questions
(b) Storage, gzipped and indexed
VCF route
RDF route
Size on disk
24.7×
60.5×
20.7×
57.5×
3.1 MB
75.8 MB
3.5 MB
214.5 MB
139.3 MB
2.9 GB
140.9 MB
8.1 GB
gzip
indexed
gzip
indexed
100,000 HG005 records
Complete HG005 VCF
(c) Regional queries by window size
Median time per query (ms)
653
118
31.5
10.3
QLever
fastest
Window size (median records per window)
(d) Parsing relative to querying, no setup
VCF parsing
faster
RDF querying
faster
1 kb windows
100 kb windows
1 Mb windows
10 Mb windows
Whole file,
per question
Whole file,
one-pass batch
2.5–2,292×
0.7×
VCF parsing time ÷ QLever time